

Optimal Adaptation under Fiscal Capacity Constraints: Doom and Virtuous Loops in Debt Markets

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Abstract

Physical climate risks such as droughts, floods, and extreme heat are increasingly recognized as systematic risk factors for the real economy and financial markets, yet we know comparatively little about whether and how public policy can reshape their pricing.

Keywords: Physical climate risks, asset pricing, public policy

1. Introduction

Many countries face a policy dilemma: physical climate risk raises expected losses and risk premia in sovereign debt markets, yet the very increase in borrowing costs

can compress fiscal space and limit investment in adaptation. This feedback has been described as a “climate–sovereign doom loop.” While conceptually compelling, the conditions under which the feedback becomes self-reinforcing—and when it can instead turn into a virtuous cycle—remain insufficiently characterized and empirically operationalized.

This paper develops a tractable macro-finance framework in which the government chooses adaptation investment G_t to manage physical climate risk and financing conditions. Adaptation reduces the persistence and volatility of climate risk and lowers both sovereign and corporate default intensities, compressing public and private spreads. We then introduce an explicit fiscal-capacity (market-access) constraint that endogenizes the shadow price of fiscal tightness, μ_t . This constraint modifies the optimality condition for G_t through a fiscal-space wedge, creating the possibility of non-linear feedback.

Our key theoretical contribution is a set of transparent threshold inequalities that classify the economy into doom-loop versus virtuous-cycle regions. The regime boundary is summarized by $\theta_t \equiv B_t(-\partial s_t^g/\partial G_t)$: when $\theta_t < 1$, a binding constraint raises the effective marginal cost of adaptation, so fiscal tightening reduces G_t^* and reinforces high spreads (doom loop); when $\theta_t > 1$, adaptation improves debt dynamics ($\partial B_{t+1}/\partial G_t < 0$), so fiscal tightness reinforces adaptation incentives (virtuous cycle). A complementary index $\rho_t \equiv \Phi_{sg}(-\partial s_t^g/\partial G_t) + \Psi_{\bar{s}^c}(-\partial \bar{s}_t^c/\partial G_t)$ captures when adaptation is self-financing on impact. We summarize the regimes in a two-panel map that clarifies policy levers to avoid the doom loop and highlights conditions under which tipping/multiplicity can arise.

Empirically, we bring these predictions to the data using a global panel of sovereign spreads (and CDS where available), public debt, and measures of adaptation/resilience. We estimate how spreads respond to adaptation capacity and interact this response with debt levels and indicators of fiscal tightness to construct empirical counterparts of θ_t and ρ_t . Consistent with the model, we find stronger non-linearities when fiscal capacity is tight, and we provide evidence that the corporate channel strengthens spread compression and expands the virtuous region.

Our results contribute to the sovereign-risk and climate-finance literatures by (i) endogenizing adaptation as an optimal policy choice under fiscal constraints, (ii) offering a regime-classification framework with empirically implementable thresholds, and (iii) highlighting a public–private complementarity channel through which adaptation can generate large fiscal returns.

2. Model Setup

2.1. Economic Environment, Climate Risk, and Adaptation

Time is discrete, $t = 0, 1, 2, \dots$. Uncertainty is generated by a climate risk factor X_t and other aggregate shocks. Government adaptation investment is denoted by $G_t \geq 0$, which can be interpreted as the flow of public spending on climate adaptation (e.g., large-scale water projects, flood defenses). For now, we treat G_t as exogenous and focus on its asset-pricing implications.

Climate risk dynamics. We model the climate factor X_t as a (possibly persistent) shock that captures the intensity of physical climate risk (e.g., drought risk) at time

t . Its dynamics are given by

$$X_{t+1} = (1 - \phi(G_t))X_t + \sigma(G_t) \varepsilon_{t+1}, \quad (1)$$

where ε_{t+1} is an i.i.d. innovation with mean zero and unit variance, $\phi(G_t) \in [0, 1)$ measures the mean-reversion of the climate shock, and $\sigma(G_t) > 0$ is the conditional volatility of X_{t+1} .

We assume:

$$\phi'(G) > 0, \quad \sigma'(G) < 0. \quad (2)$$

That is, a higher level of adaptation investment G increases the mean-reversion of climate shocks and reduces their conditional variance. Under (1)–(2), G reduces both the persistence and volatility of climate risk.

Aggregate output and tax base. Aggregate output next period is given by

$$Y_{t+1} = \bar{Y} - d_X X_{t+1} + u_{t+1}, \quad (3)$$

where $d_X > 0$ captures the adverse impact of climate risk on output and u_{t+1} is an i.i.d. disturbance. The government collects proportional taxes

$$T_{t+1} = \tau Y_{t+1}, \quad \tau \in (0, 1).$$

Stochastic discount factor. Asset prices are determined by a stochastic discount factor (SDF) M_{t+1} . We do not commit to a particular preference specification but

assume that climate risk is *bad* in the sense that

$$\text{Cov}_t(M_{t+1}, X_{t+1}) > 0. \quad (4)$$

Intuitively, high realizations of X_{t+1} (more severe climate risk) are associated with high marginal utility and thus high M_{t+1} .

2.2. Government Sector and Sovereign Debt Pricing

The government issues one-period defaultable bonds. Let B_t denote the stock of outstanding government debt at time t , and let r_t^f be the risk-free short rate. The government budget constraint is

$$B_{t+1} = (1 + r_t^g)B_t + G_t + I_t - T_t, \quad (5)$$

where r_t^g is the required return on sovereign bonds, G_t is adaptation spending, and I_t collects other government expenditures.

Default intensity. We model sovereign default in reduced form. Let $D_{t+1}^g \in \{0, 1\}$ be the indicator that the government defaults between t and $t + 1$. The (physical) default intensity is

$$\lambda_t^g = \Lambda^g(X_t, B_t, G_t), \quad (6)$$

where $\Lambda^g(\cdot)$ is increasing in climate risk and debt, but decreasing in adaptation:

$$\frac{\partial \Lambda^g}{\partial X} > 0, \quad \frac{\partial \Lambda^g}{\partial B} > 0, \quad \frac{\partial \Lambda^g}{\partial G} < 0. \quad (7)$$

These assumptions capture the idea that severe climate risk and high debt raise sovereign default risk, while greater adaptation investment improves macroeconomic resilience and reduces default intensity.

Sovereign bond pricing and spreads. Consider a one-period sovereign bond with face value one and price P_t^g . Its payoff is $(1 - D_{t+1}^g)$ in units of the numeraire. The no-arbitrage pricing equation is

$$P_t^g = \mathbb{E}_t \left[M_{t+1} (1 - D_{t+1}^g) \right]. \quad (8)$$

Define the gross required return

$$1 + r_t^g := \frac{1}{P_t^g}.$$

Let $P_t^f = \mathbb{E}_t[M_{t+1}]$ denote the price of a default-free one-period bond; its return satisfies $1 + r_t^f = 1/P_t^f$. The *sovereign spread* is defined as

$$s_t^g := r_t^g - r_t^f. \quad (9)$$

Under a standard linearization of (8), the sovereign spread can be decomposed into an expected-loss component and a risk-premium component linked to climate risk.¹

Proposition 1 (Decomposition of sovereign spreads) *Under mild regularity con-*

¹The exact functional form is not central for our comparative statics. A formal derivation is provided in the Appendix XX.

ditions and a local linearization around the mean, the sovereign spread can be written as

$$s_t^g \approx \underbrace{\lambda_t^{g,\mathbb{P}}}_{\text{expected loss}} + \underbrace{\beta_X^g(G_t) \lambda_X(G_t)}_{\text{climate risk premium}} + (\text{other risk premia}), \quad (10)$$

where $\lambda_t^{g,\mathbb{P}}$ is the physical default intensity, $\beta_X^g(G_t)$ is the sovereign bond's exposure (beta) to the climate factor X_{t+1} , and $\lambda_X(G_t)$ is the market price of climate risk. Both $\beta_X^g(G)$ and $\lambda_X(G)$ are functions of the level of adaptation G through the climate dynamics (1)–(2) and the SDF covariance (4).

We next characterize how an increase in adaptation G_t affects the sovereign spread.

Proposition 2 (Climate adaptation and sovereign financing costs) *Suppose that:*

1. Λ^g satisfies the monotonicity conditions (7);
2. G affects the climate dynamics through (1)–(2);
3. the SDF satisfies (4).

Then, holding (X_t, B_t) fixed, the sovereign spread s_t^g is decreasing in G_t :

$$\frac{\partial s_t^g}{\partial G_t} < 0. \quad (11)$$

The total effect can be decomposed into:

$$\frac{\partial s_t^g}{\partial G_t} = \underbrace{\frac{\partial \lambda_t^{g,\mathbb{P}}}{\partial G_t}}_{<0, \text{ expected loss effect}} + \underbrace{\frac{\partial \beta_X^g(G_t)}{\partial G_t} \lambda_X(G_t) + \beta_X^g(G_t) \frac{\partial \lambda_X(G_t)}{\partial G_t}}_{\leq 0, \text{ risk premium effect}}.$$

Proof sketch. The expected-loss effect follows directly from (6)–(7). The risk-premium effect arises because G reduces both the variance of X_{t+1} and its persistence, thereby lowering the sovereign bond’s exposure $\beta_X^g(G)$, and because the market price of climate risk $\lambda_X(G)$ is decreasing in the volatility of X_{t+1} given (4). A formal derivation is provided in Appendix XX. ■

Corollary 1 (Cross-sectional heterogeneity in sovereign spread relief) *The impact of adaptation investment G_t on sovereign spreads is stronger when:*

1. *the economy is more climate-exposed (higher d_X in (3)), and/or*
2. *the government debt level B_t is higher (so that a given reduction in default intensity generates a larger change in expected loss).*

2.3. Corporate Sector and Credit Spreads

We now introduce a continuum of firms indexed by i . Each firm issues defaultable corporate bonds and is exposed to both climate risk and sovereign risk.

Corporate cash flows and climate exposure. Firm i ’s cash flow next period is

$$CF_{i,t+1} = \bar{C}F_i - \theta_i X_{t+1} - \eta_i \lambda_t^g + \epsilon_{i,t+1}, \quad (12)$$

where $\theta_i \geq 0$ measures the firm’s technological exposure to climate risk (e.g., water-intensive production), $\eta_i \geq 0$ captures its exposure to sovereign risk (e.g., dependence on fiscal transfers or regulated tariffs), and $\epsilon_{i,t+1}$ is an idiosyncratic shock.

Corporate default intensity. Firm i 's (physical) default intensity is modeled as

$$\lambda_{i,t}^c = \Lambda^c(X_t, G_t; Z_i), \quad (13)$$

where Z_i collects firm-level characteristics. We assume

$$\frac{\partial \Lambda^c}{\partial X} > 0, \quad \frac{\partial \Lambda^c}{\partial G} < 0, \quad (14)$$

so that worse climate conditions raise default risk, while adaptation investment attenuates it.

Corporate bond pricing and spreads. Consider a one-period zero-coupon bond issued by firm i with price $P_{i,t}^c$ and payoff $(1 - D_{i,t+1}^c)$. The price satisfies

$$P_{i,t}^c = \mathbb{E}_t \left[M_{t+1} (1 - D_{i,t+1}^c) \right], \quad (15)$$

where $D_{i,t+1}^c$ is the default indicator. The corporate bond return is $1 + r_{i,t}^c := 1/P_{i,t}^c$, and the corporate credit spread is

$$s_{i,t}^c := r_{i,t}^c - r_t^f. \quad (16)$$

Parallel to Proposition 1 and Proposition 2, we obtain:

Proposition 3 (Decomposition of corporate credit spreads) *Under a local lin-*

earization, the corporate credit spread can be written as

$$s_{i,t}^c \approx \underbrace{\lambda_{i,t}^{c,\mathbb{P}}}_{\text{expected loss}} + \underbrace{\beta_{i,X}^c(G_t) \lambda_X(G_t)}_{\text{climate risk premium}} + (\text{other risk premia}), \quad (17)$$

where $\lambda_{i,t}^{c,\mathbb{P}}$ is the firm's physical default intensity, $\beta_{i,X}^c(G_t)$ is the firm's bond exposure to climate risk, and $\lambda_X(G_t)$ is the same market price of climate risk as in (10).

Proposition 4 (Climate adaptation and corporate financing costs) *If (14) holds and G affects the climate dynamics as in (1)–(2), then, holding (X_t, Z_i) fixed, the corporate credit spread $s_{i,t}^c$ is decreasing in adaptation investment:*

$$\frac{\partial s_{i,t}^c}{\partial G_t} < 0. \quad (18)$$

The reduction in spreads is stronger for firms with larger climate-exposure coefficients θ_i and sovereign-risk loadings η_i in (12), as summarized in Corollary 2.

Corollary 2 (Cross-sectional predictions for corporate spreads) *For a given change in adaptation investment G_t , the decline in credit spreads $s_{i,t}^c$ is predicted to be:*

1. *larger for firms with higher θ_i (more climate-sensitive cash flows, e.g., water-intensive industries); and*
2. *larger for firms with higher η_i (stronger dependence on sovereign risk, e.g., state-owned enterprises or regulated utilities).*

These cross-sectional predictions provide testable implications for the empirical analysis.

2.4. Joint Government–Corporate Financing Conditions

Finally, define the average sovereign spread $\bar{s}_t^g := \mathbb{E}[s_t^g]$ and an aggregate corporate spread $\bar{s}_t^c := \mathbb{E}_i[s_{i,t}^c]$. Combining Propositions 2 and 4, we obtain:

Proposition 5 (Adaptation investment and aggregate financing costs) *Under the above assumptions, an increase in adaptation investment G_t reduces both sovereign and corporate spreads at the aggregate level:*

$$\frac{\partial \bar{s}_t^g}{\partial G_t} < 0, \quad \frac{\partial \bar{s}_t^c}{\partial G_t} < 0. \quad (19)$$

Moreover, the joint reduction in public and private financing costs can offset part of the contemporaneous fiscal cost of adaptation G_t in the government budget constraint (5).

3. Optimal Adaptation Policy

3.1. Government objective and resource constraint

In the previous subsections, government adaptation investment G_t is taken as exogenous. We now endogenize G_t by specifying a multi-objective government problem in which the policy maker trades off three goals: (i) sustaining aggregate economic activity, (ii) mitigating climate damages, and (iii) improving sovereign and corporate

financing conditions.

Let C_t denote aggregate (private) consumption and let $D(X_t)$ capture the flow disutility from climate risk X_t , with $D'(X) > 0$ and $D''(X) \geq 0$.² We also let high sovereign and corporate spreads generate additional welfare costs, as they tighten public and private financing constraints. The per-period social welfare is

$$W_t = U(C_t) - \alpha D(X_t) - \gamma^g s_t^g B_t - \gamma^c \bar{s}_t^c K_t, \quad (20)$$

where, $U(\cdot)$ is a standard increasing, concave utility function, $U'(C) > 0$, $U''(C) < 0$; $\alpha > 0$ measures the weight on reducing climate damages; $\gamma^g > 0$ and $\gamma^c > 0$ measure the social costs of high sovereign and corporate financing spreads, respectively; s_t^g is the sovereign spread defined in (9), B_t is government debt; $\bar{s}_t^c$ is an aggregate corporate spread defined in (17); K_t measures the aggregate scale of private borrowing or capital.

The multi-objective nature of the problem is made explicit in (20): the government puts (implicit) weights on consumption, climate damages, and financing conditions. In practice, this is equivalent to a standard single-objective optimization with a weighted composite criterion.

For concreteness, we assume that aggregate consumption is given by

$$C_t = Y_t - G_t - \Phi(B_t, s_t^g) - \Psi(K_t, \bar{s}_t^c), \quad (21)$$

where Y_t is aggregate output defined in (3), G_t is adaptation spending, and $\Phi(\cdot)$

²Modeling climate damages partly as utility/amenity damages is standard in the climate-economy literature; see, e.g., Barrage (2020).

and $\Psi(\cdot)$ capture government and private resources absorbed by debt service and financing frictions, with

$$\frac{\partial \Phi}{\partial s^g} > 0, \quad \frac{\partial \Psi}{\partial s^c} > 0.$$

Thus, higher sovereign and corporate spreads tighten the resource constraint by increasing required payments and reducing available consumption.

Government debt evolves according to the budget constraint (5), repeated here for convenience:

$$B_{t+1} = (1 + r_t^g)B_t + G_t + I_t - T_t, \quad (22)$$

with $r_t^g = r_t^f + s_t^g$.

3.2. FOC and characterization of G_t^*

The government chooses a sequence $\{G_t\}_{t \geq 0}$ to maximize the expected discounted sum of social welfare:

$$\max_{\{G_t\}_{t \geq 0}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t W_t, \quad \beta \in (0, 1), \quad (23)$$

subject to the climate dynamics (1), the output equation (3), the government budget constraint (22), the bond pricing relations (8) and (15), and the definitions of sovereign and corporate spreads in (9) and (17).

Let the state vector be $S_t = (X_t, B_t, \Xi_t)$, where Ξ_t collects other relevant aggregate states (e.g., K_t and exogenous macro shocks). The government's value function

satisfies

$$V(S_t) = \max_{G_t \geq 0} \left\{ W_t(S_t, G_t) + \beta \mathbb{E}_t[V(S_{t+1})] \right\}, \quad (24)$$

where S_{t+1} evolves according to the laws of motion implied by the constraints and the climate dynamics (1).

Consider an interior solution and differentiability. The first-order condition (FOC) for G_t can be written as

$$\frac{\partial W_t}{\partial G_t} + \beta \mathbb{E}_t \left[V_X(S_{t+1}) \frac{\partial X_{t+1}}{\partial G_t} + V_B(S_{t+1}) \frac{\partial B_{t+1}}{\partial G_t} + V_\Xi(S_{t+1}) \frac{\partial \Xi_{t+1}}{\partial G_t} \right] = 0. \quad (25)$$

At time t , (X_t, B_t, K_t, Ξ_t) are predetermined; contemporaneous effects of G_t operate through resource allocation and through prices (spreads) embedded in $\Phi(\cdot)$ and $\Psi(\cdot)$. Using (20)–(21), the marginal effect of G_t on current welfare can be expressed as

$$\frac{\partial W_t}{\partial G_t} = U_C(C_t) \frac{\partial C_t}{\partial G_t} - \gamma^g B_t \frac{\partial s_t^g}{\partial G_t} - \gamma^c K_t \frac{\partial \bar{s}_t^c}{\partial G_t}, \quad (26)$$

where

$$\frac{\partial C_t}{\partial G_t} = -1 + \Phi_{sg}(B_t, s_t^g) \left(-\frac{\partial s_t^g}{\partial G_t} \right) + \Psi_{\bar{s}^c}(K_t, \bar{s}_t^c) \left(-\frac{\partial \bar{s}_t^c}{\partial G_t} \right). \quad (27)$$

Equation (27) makes clear that the direct resource cost of adaptation (-1) can be partially offset by contemporaneous spread compression that releases public and private resources.

Combining (25)–(27), the optimality condition equates marginal cost and marginal benefits. We summarize this characterization as follows.

Proposition 6 (Characterization of optimal adaptation) *Suppose (2) and (7)*

hold and, analogously, adaptation reduces aggregate corporate spreads so that $\partial \bar{s}_t^c / \partial G_t <$

0. Then the government's optimal adaptation policy G_t^* satisfies

$$\begin{aligned}
U_C(C_t) = & \underbrace{U_C(C_t) \left[\Phi_{sg}(B_t, s_t^g) \left(-\frac{\partial s_t^g}{\partial G_t} \right) + \Psi_{\bar{s}^c}(K_t, \bar{s}_t^c) \left(-\frac{\partial \bar{s}_t^c}{\partial G_t} \right) \right]}_{\text{current benefit: resource release via spread compression}} + \underbrace{\gamma^g B_t \left(-\frac{\partial s_t^g}{\partial G_t} \right) + \gamma^c K_t \left(-\frac{\partial \bar{s}_t^c}{\partial G_t} \right)}_{\text{current benefit: lower financing costs}} \\
& + \underbrace{\beta \mathbb{E}_t \left[V_X(S_{t+1}) \frac{\partial X_{t+1}}{\partial G_t} + V_B(S_{t+1}) \frac{\partial B_{t+1}}{\partial G_t} + V_\Xi(S_{t+1}) \frac{\partial \Xi_{t+1}}{\partial G_t} \right]}_{\text{intertemporal benefit term}}. \quad (28)
\end{aligned}$$

In particular, optimal adaptation trades off the direct resource cost against (i) contemporaneous fiscal returns from spread compression and lower financing costs, and (ii) discounted benefits through next-period states, including future damage mitigation via $\partial X_{t+1} / \partial G_t$ and debt-sustainability effects via $\partial B_{t+1} / \partial G_t$.

Proposition 6 highlights that even before introducing an explicit fiscal-capacity constraint, adaptation can be partially self-financing through contemporaneous credit-market improvements. This feature will be central in the regime identification below.

3.3. Fiscal Capacity Constraint and Regime Identification

3.3.1. Fiscal capacity constraint

To capture limited market access and fiscal space, we introduce an explicit fiscal-capacity constraint on next-period debt:

$$B_{t+1} \leq \bar{B}(\Xi_t), \quad (29)$$

where $\bar{B}(\Xi_t)$ is an upper bound that depends on sovereign fundamentals Ξ_t (e.g., institutional quality, resilience, external buffers). Let $\mu_t \geq 0$ denote the associated multiplier. When the constraint binds ($\mu_t > 0$), fiscal tightness has a shadow value, and optimal adaptation is distorted relative to the unconstrained benchmark.

3.3.2. Self-financing and debt-improving indices

The constrained problem modifies the optimality condition through a fiscal-space wedge that depends on how adaptation affects next-period debt. A key object is the marginal effect of G_t on debt dynamics:

$$\frac{\partial B_{t+1}}{\partial G_t} = 1 + B_t \frac{\partial s_t^g}{\partial G_t}. \quad (30)$$

Motivated by (27) and (30), we define two indices:

$$\rho_t \equiv \Phi_{sg}(B_t, s_t^g) \left(-\frac{\partial s_t^g}{\partial G_t} \right) + \Psi_{\bar{s}^c}(K_t, \bar{s}_t^c) \left(-\frac{\partial \bar{s}_t^c}{\partial G_t} \right), \quad \theta_t \equiv B_t \left(-\frac{\partial s_t^g}{\partial G_t} \right). \quad (31)$$

The first index ρ_t captures whether adaptation is self-financing on impact through public and private resource release, while the second index θ_t captures whether spread compression is strong enough to improve debt dynamics.

Corollary 3 (Self-financing on impact) *If $\rho_t \geq 1$, then the contemporaneous resource effect satisfies $\partial C_t / \partial G_t \geq 0$ in (27): adaptation is weakly self-financing on impact through reduced public and private resources absorbed by spreads.*

Corollary 4 (Debt-improving adaptation) *If $\theta_t > 1$, then $\partial B_{t+1}/\partial G_t < 0$ in (30): increasing adaptation reduces next-period debt by lowering sovereign spreads sufficiently strongly.*

3.3.3. Doom versus virtuous cycles and tipping

When the fiscal-capacity constraint binds ($\mu_t > 0$), the sign of $\partial B_{t+1}/\partial G_t$ determines whether fiscal tightness acts as a deterrent or an additional incentive to invest in adaptation. Intuitively, if $\theta_t < 1$, increasing G_t raises B_{t+1} mechanically and does not generate enough spread relief to offset the debt effect, so the constraint raises the effective marginal cost of G_t . If $\theta_t > 1$, adaptation reduces B_{t+1} ; a binding constraint then strengthens the incentive to invest in G_t .

Proposition 7 (Doom vs. virtuous loop under a fiscal-capacity constraint)

Consider the constraint (29) with multiplier $\mu_t \geq 0$ and define θ_t as in (31). When the constraint binds ($\mu_t > 0$), the economy exhibits: (i) a doom-loop region if $\theta_t < 1$, in which tighter fiscal capacity reduces G_t^ and reinforces high spreads; (ii) a virtuous-cycle region if $\theta_t > 1$, in which adaptation improves debt dynamics and fiscal tightness strengthens incentives to invest in G_t . Moreover, tipping/multiplicity is more likely in a neighborhood of $\theta_t \approx 1$ when the feedback between fiscal tightness and policy is sufficiently steep (formalized in Corollary 5).*

Corollary 5 (Tipping and local multiplicity) *Let the constrained best response be summarized by $G_t^* = \mathcal{G}(\mu_t; S_t)$ and fiscal tightness by $\mu_t = \mathcal{M}(B_{t+1} - \bar{B}(\Xi_t))$ locally*

around the boundary, with $\mathcal{M}'(\cdot) > 0$. If the local feedback satisfies

$$|\partial \mathcal{G}(\mu_t; S_t)/\partial \mu_t| |\mathcal{M}'(\cdot)| |1 + B_t(\partial s_t^g/\partial G_t)| > 1, \quad (32)$$

then small changes in fiscal tightness can trigger large changes in optimal adaptation (tipping), and the model can exhibit locally multiple equilibria.

Figure T.1 summarizes the identification logic in a two-panel regime map. Panel A ($\mu_t = 0$) highlights the role of ρ_t for the contemporaneous fiscal return to adaptation. Panel B ($\mu_t > 0$) highlights the doom versus virtuous boundary at $\theta_t = 1$, with a narrow band around $\theta_t \approx 1$ illustrating heightened tipping risk.

3.3.4. Corporate channel and public–private complementarity

The corporate channel expands the virtuous region through two effects captured by (31). First, by reducing aggregate corporate spreads, adaptation releases private resources absorbed by financing frictions, increasing ρ_t through $\Psi_{\bar{s}^c}(-\partial \bar{s}_t^c/\partial G_t)$. Second, improved private financial conditions can feed back into sovereign risk, strengthening the sovereign spread response $-\partial s_t^g/\partial G_t$ and increasing θ_t . In the regime map, these effects shift the economy upward and to the right, making self-financing on impact more likely and reducing the set of states in which fiscal tightness triggers a doom loop.

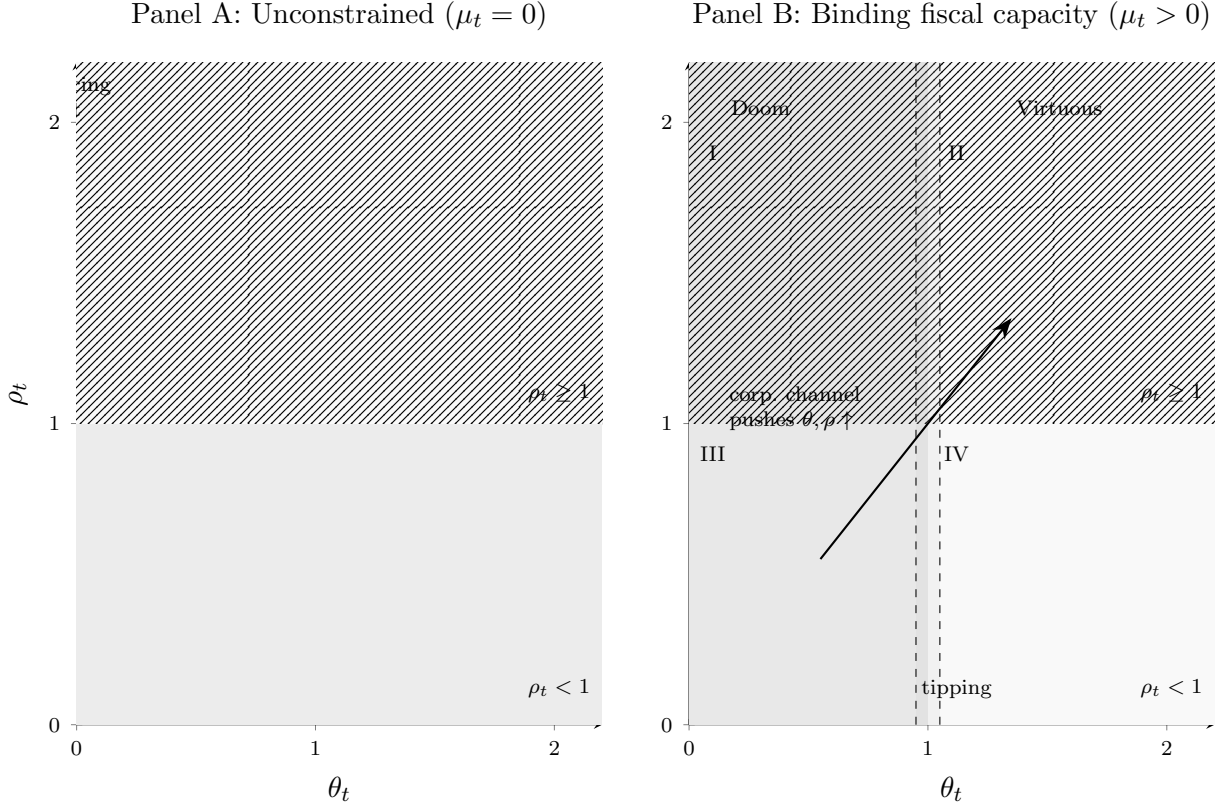


Figure T.1: Regime map accompanying Proposition 7. Panel A ($\mu_t = 0$) emphasizes the self-financing threshold $\rho_t = 1$. Panel B ($\mu_t > 0$) highlights the doom vs. virtuous boundary at $\theta_t = 1$; the shaded band around $\theta_t \approx 1$ is illustrative of where tipping/multiplicity is more likely under strong feedback (Corollary 6). In Panel B, quadrants I–IV follow the standard convention (upper-left, upper-right, lower-left, lower-right).

4. Empirical Evidence

This section provides reduced-form evidence for the model’s main mechanism. I first test whether adaptation readiness is associated with lower sovereign spreads. I then construct an empirical measure of spread relief and ask whether it sorts debt dynamics into debt-worsening and debt-improving regions. The evidence is inter-

puted as mechanism-consistent rather than causal, because the adaptation proxy captures broad readiness rather than directly observed public adaptation outlays.

4.1. Data, Proxies, and Measurement Units

The sample is an unbalanced country-year panel covering 1995–2023. The United States is excluded because sovereign spreads are measured relative to the U.S. benchmark. The main spread outcome is s_{it} , and debt exposure is debt/GDP, B_{it} .

The two climate proxies come from the GDP-adjusted ND-GAIN measures. I denote GDP-adjusted readiness by G_{it} and GDP-adjusted vulnerability-performance by X_{it} ; both compare a country’s observed score with the score predicted by GDP per capita in the same year. I use these adjusted measures to separate relative adaptation performance from the broader level of economic development. Higher values of both proxies indicate better performance: higher readiness relative to GDP for G_{it} , and lower vulnerability relative to GDP for X_{it} .

All ratio and rate variables are kept in share units, so 0.01 corresponds to one percentage point. The control vector $\mathbf{Z}_{it}$ includes standard macroeconomic, fiscal, external-liquidity, institutional, and terms-of-trade controls. The main specifications include country and year fixed effects. Robust t -statistics are reported in parentheses. Table 1 reports descriptive statistics.

Table 1: Post-Preprocessing Descriptive Statistics

Block	Variable	N	Mean	SD	Median	Min.	Max.
S	Sovereign spread	1,272	0.024	0.043	0.007	-0.034	0.343
G	GDP-adj. readiness performance	1,798	0.058	0.089	0.048	-0.326	0.303
B	Debt/GDP	1,712	0.586	0.342	0.522	0.039	2.610
X	GDP-adj. vulnerability performance	1,798	-0.040	0.058	-0.047	-0.185	0.295
Z_1	Real GDP	1,791	0.080	0.029	0.077	0.019	0.163
Z_2	Real GDP growth	1,787	0.034	0.036	0.035	-0.145	0.246
Z_3	CPI inflation	1,791	0.057	0.108	0.032	-0.018	1.973
Z_4	Overall balance/GDP	1,697	-0.004	0.034	-0.005	-0.300	0.234
Z_5	International reserves	1,753	0.055	0.128	0.018	0.000	1.527
Z_6	Government effectiveness	1,550	0.006	0.009	0.007	-0.013	0.025
Z_7	Regulatory quality	1,550	0.006	0.008	0.007	-0.013	0.023
Z_8	Terms of trade	1,595	1.006	0.187	0.994	0.319	2.731

Notes: Statistics use the post-preprocessing non-U.S. 1995–2023 panel. Each row is computed on nonmissing observations for the corresponding variable.

4.2. Adaptation and Sovereign Financing Costs

The baseline spread regression tests the sign prediction in Proposition 2:

$$s_{it} = \alpha_i + \tau_t + \beta_G G_{it} + \beta_B B_{it} + \beta_X X_{it} + \mathbf{\Gamma}' \mathbf{Z}_{it} + u_{it}. \quad (33)$$

Table 2 reports the baseline fixed-effects estimate. The coefficient on G_{it} is negative and statistically significant: countries with stronger readiness performance face lower sovereign spreads. Debt/GDP enters positively, as in standard sovereign-risk regressions. The evidence is consistent with Proposition 2, subject to the reduced-form interpretation of the proxy.

Table 2: Baseline Fixed-Effects Estimates

Dependent variable	10-year sovereign spread, s_{it}
G_{it}	-0.046*** (-2.804)
B_{it}	0.043*** (6.042)
X_{it}	-0.087*** (-2.626)
Controls $\mathbf{Z}_{it}$	Yes
Country FE	Yes
Year FE	Yes
Sample period	1995–2023
Countries	60
Observations	1,120
Adjusted R^2	0.900

Notes: Robust t -statistics are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

4.3. Heterogeneous Spread Relief and the Empirical Debt-Improving Index

The next step asks whether the spread-relief effect of the GDP-adjusted readiness-performance proxy varies with debt exposure and GDP-adjusted vulnerability performance, as suggested by Corollary 1. The full-interaction spread model is

$$s_{it} = \alpha_i + \tau_t + \beta_G G_{it} + \beta_B B_{it} + \beta_X X_{it} + \beta_{GB}(G_{it} \times B_{it}) + \beta_{GX}(G_{it} \times X_{it}) + \mathbf{\Gamma}' \mathbf{Z}_{it} + u_{it}. \quad (34)$$

For the empirical index construction, the expanded full-interaction specification also allows the marginal effect of G_{it} to vary with the control vector:

$$\begin{aligned} s_{it} = & \alpha_i + \tau_t + \beta_G G_{it} + \beta_B B_{it} + \beta_X X_{it} \\ & + \beta_{GB}(G_{it} \times B_{it}) + \beta_{GX}(G_{it} \times X_{it}) + \mathbf{\Pi}'(G_{it} \times \mathbf{Z}_{it}) + \mathbf{\Gamma}' \mathbf{Z}_{it} + u_{it}. \end{aligned} \quad (35)$$

Here, $G_{it} \times \mathbf{Z}_{it}$ denotes the vector of interactions between G_{it} and each component of $\mathbf{Z}_{it}$. The implied marginal effect of G_{it} in equation (35) is

$$\frac{\partial s_{it}}{\partial G_{it}} = \beta_G + \beta_{GB}B_{it} + \beta_{GX}X_{it} + \mathbf{\Pi}'\mathbf{Z}_{it}. \quad (36)$$

The implied marginal spread relief and empirical debt-improving index are

$$\widehat{m}_{it}^{G,F} = -\frac{\partial \widehat{s}_{it}}{\partial G_{it}}, \quad \widehat{\theta}_{it}^F = B_{it}\widehat{m}_{it}^{G,F}. \quad (37)$$

The index $\widehat{\theta}_{it}^F$ should be interpreted as an empirical proxy for debt-improving capacity, not as a structural estimate of the theoretical θ_t .

Table 3 reports the interaction estimates. Spread relief from G_{it} is stronger at higher debt levels, as shown by the negative $G_{it} \times B_{it}$ coefficient. The full-interaction columns also allow relief to vary with vulnerability performance and the control vector. Table 4 summarizes the resulting index, $\widehat{\theta}_{it}^F = B_{it}\widehat{m}_{it}^{G,F}$. This is a proxy-scaled sorting variable, not a structural estimate of the theoretical threshold $\theta_t = 1$.

Table 3: Heterogeneous Spread Relief

Dependent variable	10-year sovereign spread, s_{it}			
G_{it}	0.053** (2.116)	-0.046*** (-2.791)	0.058** (2.271)	0.319*** (4.651)
B_{it}	0.062*** (6.922)	0.044*** (6.056)	0.064*** (6.995)	0.070*** (7.456)
X_{it}	-0.052 (-1.562)	-0.084** (-2.485)	-0.043 (-1.244)	0.086* (1.900)
$G_{it} \times B_{it}$	-0.211*** (-4.795)		-0.221*** (-4.892)	-0.266*** (-5.723)
$G_{it} \times X_{it}$		0.121 (1.196)	0.268** (2.302)	0.339** (2.255)
Controls $\mathbf{Z}_{it}$	Yes	Yes	Yes	Yes
$G_{it} \times \mathbf{Z}_{it}$ controls	No	No	No	Yes
Country FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Countries	60	60	60	60
Observations	1,120	1,120	1,120	1,120
Adjusted R^2	0.905	0.900	0.906	0.911

Notes: Columns 1–4 report debt heterogeneity, GDP-adjusted vulnerability-performance heterogeneity, full interaction, and full interaction with $G_{it} \times \mathbf{Z}_{it}$, respectively. Robust t -statistics are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 4: Empirical Debt-Improving Index Descriptive Statistics

Variable	N	Mean	SD	Median	Min.	Max.
$\hat{\theta}_{it}^F$	1,120	0.062	0.151	0.020	-0.094	1.285
$\hat{m}_{it}^{G,F}$	1,120	0.056	0.106	0.044	-0.184	0.562
$\partial \hat{s}_{it} / \partial G_{it}$	1,120	-0.056	0.106	-0.044	-0.562	0.184

Notes: Statistics are computed on the expanded full-interaction spread-model estimation sample. The marginal relief variable is $\hat{m}_{it}^{G,F} = -\partial \hat{s}_{it} / \partial G_{it}$, and the empirical debt-improving index is $\hat{\theta}_{it}^F = B_{it} \hat{m}_{it}^{G,F}$.

4.4. Visualizing Spread Relief and the Empirical Index

Figure 1 plots the implied marginal spread relief, $-\partial \hat{s}_{it} / \partial G_{it}$. Panel A varies debt/GDP, and Panel B varies vulnerability performance. The figure shows that the estimated relief increases with debt exposure and is larger for more vulnerable observations.

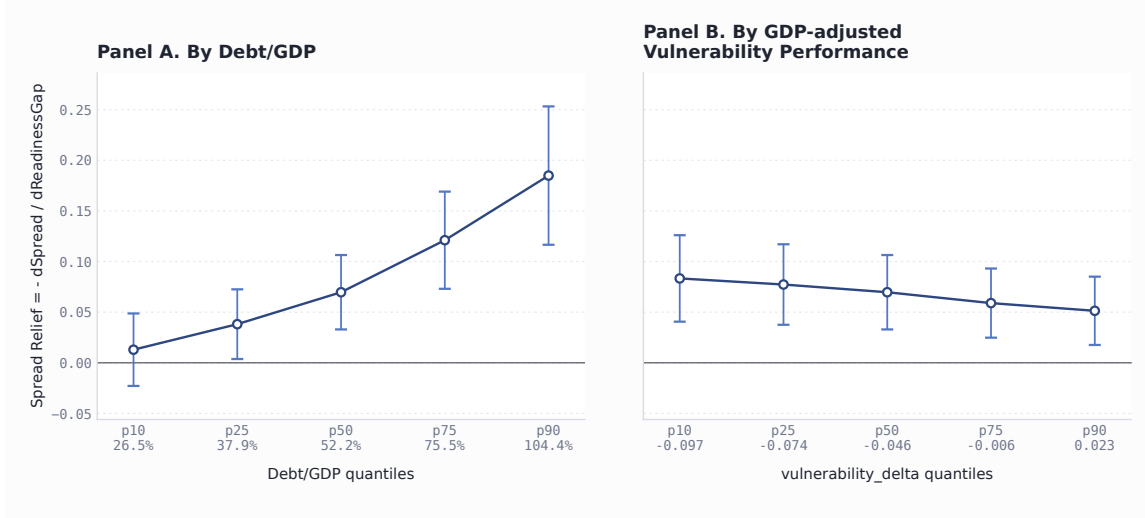


Figure 1: Marginal Spread Relief from the Full-Interaction Spread Model
Notes: Panel A fixes X_{it} at its median. Panel B fixes B_{it} at its median. Higher X_{it} indicates lower vulnerability relative to GDP-predicted vulnerability, so the downward slope in Panel B indicates stronger spread relief among more vulnerable observations.

Figure 2 plots the country-year distribution of $\hat{\theta}_{it}^F$ and country-level average rankings. The dashed line marks the preferred empirical cutoff selected by the kink specification below. It should not be interpreted as the theoretical threshold $\theta_t = 1$.

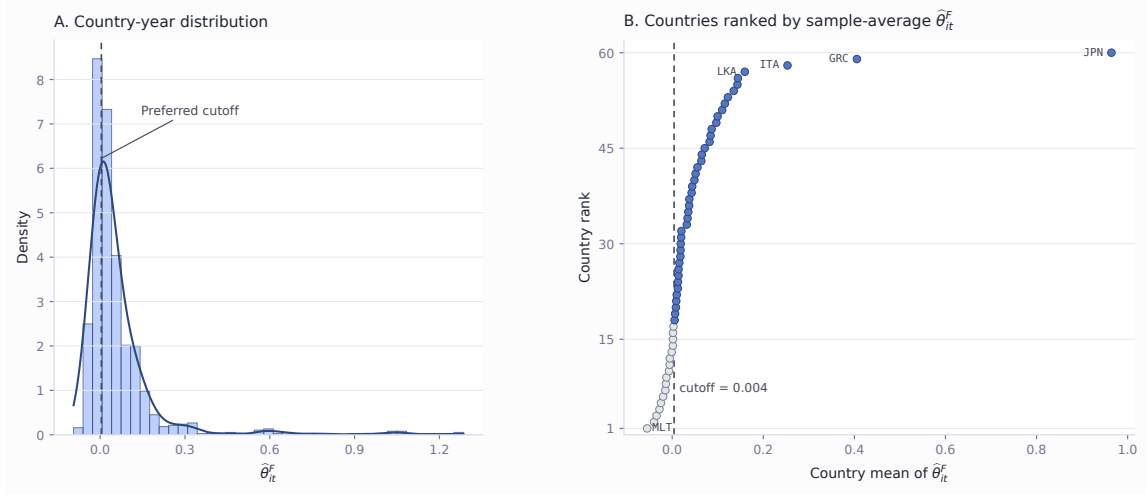


Figure 2: Distribution of the Empirical Debt-Improving Index

Notes: The left panel plots the country-year distribution of $\hat{\theta}_{it}^F$. The right panel ranks countries by sample-average $\hat{\theta}_{it}^F$. The dashed line marks the preferred empirical cutoff from the without-theta-controls kink specification.

4.5. Debt Dynamics and the Continuous Theta Interaction

The next set of tests asks whether the GDP-adjusted readiness-performance proxy is associated with next-period debt changes in a way that varies with the empirical debt-improving index. Define

$$\Delta B_{i,t+1} = B_{i,t+1} - B_{it}. \quad (38)$$

The baseline debt-change regression is

$$\Delta B_{i,t+1} = \alpha_i + \tau_t + \lambda_0 G_{it} + \mathbf{\Gamma}' \mathbf{Z}_{it} + \varepsilon_{it}, \quad (39)$$

and the continuous theta interaction model is

$$\Delta B_{i,t+1} = \alpha_i + \tau_t + \lambda_0 G_{it} + \lambda_1 (G_{it} \times \widehat{\theta}_{it}^F) + \mathbf{\Gamma}' \mathbf{Z}_{it} + \varepsilon_{it}. \quad (40)$$

Table 5 reports the continuous theta-interaction test. The coefficient on $G_{it} \times \widehat{\theta}_{it}^F$ is negative in Column 2, indicating that the debt effect of readiness declines as estimated debt-improving capacity rises. Column 3 adds the theta main effect; the interaction becomes weaker, consistent with interpreting $\widehat{\theta}_{it}^F$ primarily as a sorting variable.

Table 5: Debt-Change Regressions

Dependent variable	$\Delta B_{i,t+1} = B_{i,t+1} - B_{it}$		
G_{it}	-0.073 (-1.463)	-0.042 (-0.810)	-0.015 (-0.302)
$G_{it} \times \widehat{\theta}_{it}^F$		-0.886*** (-2.707)	-0.310 (-0.660)
$\widehat{\theta}_{it}^F$			-0.141* (-1.898)
Theta main effect	No	No	Yes
Controls $\mathbf{Z}_{it}$	Yes	Yes	Yes
Country FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Observations	1,060	1,060	1,060
Countries	60	60	60
Adjusted R^2	0.425	0.433	0.441

Notes: Columns 1–3 report the baseline model, the continuous theta-interaction model, and the specification with a theta main effect. Controls are included but not reported. Robust t -statistics are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

4.6. Grouped Debt-Change Heterogeneity

Table 6 reports quintile regressions. Sorting by the composite index $\widehat{\theta}_{it}^F$ gives the clearest pattern: the coefficient on G_{it} is positive in the lowest group and negative

in the highest groups. Debt/GDP alone produces a weaker ordering, while marginal relief alone is closer but lacks the debt-exposure component of the model.

Table 6: Grouped Debt-Change Heterogeneity Regressions

Panel A. Full-theta groups					
Dependent variable	$B_{i,t+1} - B_{it}$				
Group	0–20%	20–40%	40–60%	60–80%	80–100%
G_{it}	0.243* (1.727)	0.057 (0.866)	-0.152 (-1.550)	-0.249* (-1.770)	-0.372** (-2.077)
Controls $\mathbf{Z}_{it}$	Yes	Yes	Yes	Yes	Yes
Country FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Observations	212	212	212	212	212
Countries	25	33	35	37	22
Adjusted R^2	0.604	0.669	0.499	0.366	0.522
Panel B. Debt-ratio groups					
Dependent variable	$B_{i,t+1} - B_{it}$				
Group	0–20%	20–40%	40–60%	60–80%	80–100%
G_{it}	0.060 (0.861)	-0.116 (-1.336)	-0.067 (-1.021)	0.038 (0.291)	-0.363 (-1.317)
Controls $\mathbf{Z}_{it}$	Yes	Yes	Yes	Yes	Yes
Country FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Observations	212	212	212	212	212
Countries	28	35	41	32	22
Adjusted R^2	0.466	0.483	0.700	0.394	0.625
Panel C. Marginal-relief groups					
Dependent variable	$B_{i,t+1} - B_{it}$				
Group	0–20%	20–40%	40–60%	60–80%	80–100%
G_{it}	0.325** (2.244)	0.097 (0.732)	-0.041 (-0.426)	-0.134 (-1.350)	-0.512*** (-3.273)
Controls $\mathbf{Z}_{it}$	Yes	Yes	Yes	Yes	Yes
Country FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Observations	212	212	212	212	212
Countries	25	36	36	34	24
Adjusted R^2	0.603	0.568	0.518	0.606	0.493

Notes: Each column estimates the next-period change in debt/GDP on current GDP-adjusted readiness performance, G_{it} , within the indicated subgroup. Macro control coefficients and subgroup cutoff values are omitted for compactness. Robust t -statistics are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

4.7. Single-Crossing Kink Test

The single-crossing specification provides the closest empirical analogue to Corollary 4 and its mirror image. It directly tests whether the marginal effect of the adaptation-readiness proxy on next-period debt changes switches sign around an empirical cutoff. This specification is not a standard panel threshold model. It is designed to estimate whether the marginal debt effect of G_{it} crosses zero as a function of $\widehat{\theta}_{it}^F$.

For a candidate cutoff c , the estimated specification is

$$\Delta B_{i,t+1} = \alpha_i + \tau_t + aG_{it}(c - \widehat{\theta}_{it}^F)_+ + bG_{it}(\widehat{\theta}_{it}^F - c)_+ + \mathbf{\Gamma}'\mathbf{Z}_{it} + \varepsilon_{it}. \quad (41)$$

The implied marginal effect is

$$m(\theta; c) = a(c - \theta)_+ + b(\theta - c)_+. \quad (42)$$

The theoretical sign pattern is

$$a > 0, \quad b < 0.$$

Table 7 selects an empirical cutoff of $c = 0.004123$. The estimated marginal effect is positive below the cutoff and negative above it, matching the debt-worsening/debt-improving sign pattern. The cutoff is a proxy-scaled empirical boundary rather than the literal theoretical threshold $\theta_t = 1$.

Table 7: Single-Crossing Kink Marginal-Effect Model

Dependent variable	$B_{i,t+1} - B_{it}$
$G_{it}(c - \hat{\theta}_{it}^F)_+$	3.380*** (2.592)
$G_{it}(\hat{\theta}_{it}^F - c)_+$	-0.804** (-2.464)
Cutoff c	0.004
RSS	1.951
Share below cutoff	0.350
Selection rule	RSS minimum; sign-consistent
Controls $\mathbf{Z}_{it}$	Yes
Country FE	Yes
Year FE	Yes
Observations	1,060
Adjusted R^2	0.435

Notes: The exact RSS-selected cutoff is $c = 0.0041231435658208$. The model includes country and year fixed effects and $\mathbf{Z}_{it}$ controls, but does not include direct theta controls. Robust t -statistics are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$.

Figure 3 plots the continuous marginal-effect curve implied by equation (42). The left side of the cutoff corresponds to the debt-worsening region, while the right side corresponds to the debt-improving region.

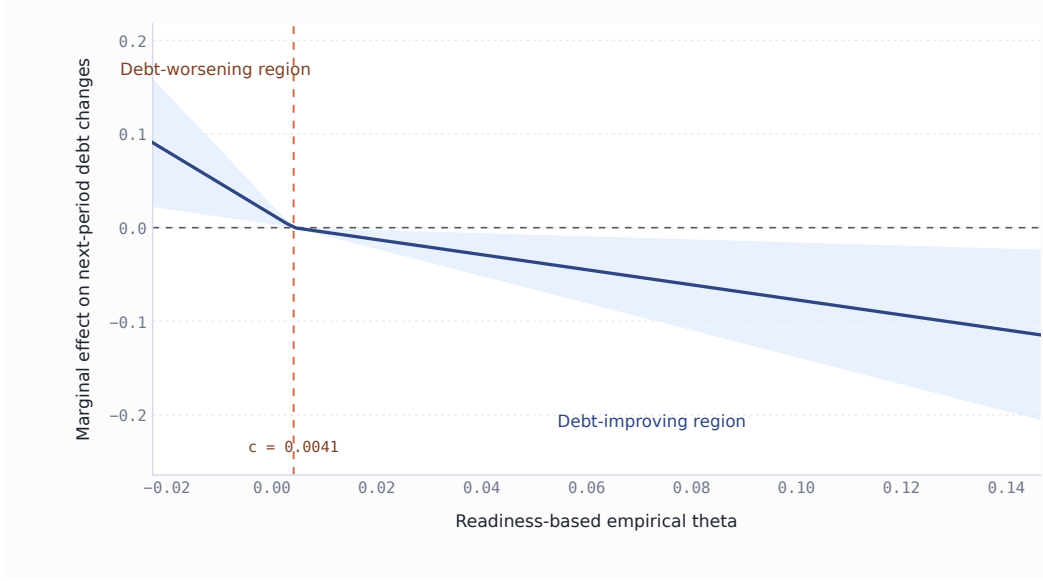


Figure 3: Continuous Kink Marginal Effect without Theta Controls

Notes: The figure plots $m(\theta; c) = a(c - \theta)_+ + b(\theta - c)_+$ from the single-crossing kink model without theta controls. Positive values imply that readiness improvements are associated with higher next-period debt changes; negative values imply lower next-period debt changes. The vertical dashed line marks the empirical cutoff c .

4.8. Country Screens Around the Preferred Cutoff

Finally, Table 8 reports an illustrative country-level screen based on the preferred cutoff. The exercise is descriptive and should be read as a diagnostic use of the index, not as a structural country ranking.

Future work can use this screen to choose representative case countries. Following the budget-document approach in Duffy’s *Climate Change, Adaptation, and Sovereign Risk*, we can use public budget files and spending reports to identify actual adaptation expenditures, providing a direct benchmark for the readiness proxy and the debt-improving index.

Panel A. Low debt, theta always above the cutoff, and top-quartile marginal relief**Table 8:** Country Screens Around the Preferred Cutoff

Country	ISO3	Sample period	Years	Mean debt/GDP	Mean theta	Mean $\hat{m}^{G,F}$
Vietnam	VNM	2007–2023	17	39.50%	0.065	0.164
Indonesia	IDN	2003–2023	21	33.72%	0.055	0.156
Colombia	COL	2002–2023	21	45.82%	0.064	0.134

Panel B. High debt, theta always below the cutoff, and bottom-quartile marginal relief

Country	ISO3	Sample period	Years	Mean debt/GDP	Mean theta	Mean $\hat{m}^{G,F}$
Malta	MLT	2010–2023	14	53.91%	-0.055	-0.107

Notes: The preferred cutoff is $c = 0.004123$. Screening thresholds use country-level means: the mean-debt median is 53.06 percent, the bottom quartile of mean $\hat{m}_{it}^{G,F}$ is 0.003, and the top quartile is 0.112. The always-above and always-below conditions are evaluated year by year.

4.9. Summary and Online Appendix Roadmap

The empirical section delivers three reduced-form patterns. First, GDP-adjusted readiness is negatively associated with sovereign spreads. Second, the implied spread relief is stronger when debt exposure is high and when countries are more vulnerable relative to their GDP-predicted benchmark. Third, the debt-dynamics results are state dependent: readiness is associated with higher next-period debt changes in low- $\hat{\theta}_{it}^F$ regions and lower debt changes in high- $\hat{\theta}_{it}^F$ regions. These findings are consistent with the paper’s central mechanism.

The online appendix reports supporting exercises. Appendix B varies the debt-dynamics specifications and shows that the without-theta-controls kink model gives the clearest single-crossing pattern. Appendix C repeats the analysis using level ND-GAIN readiness and vulnerability measures. The main readiness and debt-interaction results remain stable, while vulnerability heterogeneity is more proxy-sensitive.

References

Barrage, L. (2020). Optimal dynamic carbon taxes in a climate–economy model with distortionary fiscal policy. *Review of Economic Studies*, 87(1):1–39.

A. Proofs

A.1. Proof of Proposition 7: Doom vs. virtuous loop under a fiscal-capacity constraint

Consider the government's constrained one-period problem at date t :

$$V^c(S_t) = \max_{G_t \geq 0} \left\{ W_t(S_t, G_t) + \beta \mathbb{E}_t[V^c(S_{t+1})] + \mu_t(\bar{B}(\Xi_t) - B_{t+1}) \right\}, \quad (\text{A.1})$$

where $\mu_t \geq 0$ is the multiplier on the fiscal-capacity constraint (29), together with the complementarity conditions

$$\mu_t \geq 0, \quad \bar{B}(\Xi_t) - B_{t+1} \geq 0, \quad \mu_t(\bar{B}(\Xi_t) - B_{t+1}) = 0. \quad (\text{A.2})$$

For an interior choice of G_t , the first-order condition is

$$\frac{\partial W_t}{\partial G_t} + \beta \mathbb{E}_t \left[V_X^c(S_{t+1}) \frac{\partial X_{t+1}}{\partial G_t} + V_B^c(S_{t+1}) \frac{\partial B_{t+1}}{\partial G_t} + V_\Xi^c(S_{t+1}) \frac{\partial \Xi_{t+1}}{\partial G_t} \right] - \mu_t \frac{\partial B_{t+1}}{\partial G_t} = 0. \quad (\text{A.3})$$

Relative to the unconstrained first-order condition (25), the binding fiscal-capacity constraint adds the wedge $-\mu_t(\partial B_{t+1}/\partial G_t)$.

Using the debt law of motion in (30) and the definition of θ_t in (31),

$$\frac{\partial B_{t+1}}{\partial G_t} = 1 + B_t \frac{\partial s_t^g}{\partial G_t} = 1 - \theta_t. \quad (\text{A.4})$$

Hence (A.3) can be written as

$$H(G_t, \mu_t; S_t) \equiv \mathcal{F}(G_t; S_t) - \mu_t(1 - \theta_t) = 0, \quad (\text{A.5})$$

where

$$\mathcal{F}(G_t; S_t) \equiv \frac{\partial W_t}{\partial G_t} + \beta \mathbb{E}_t \left[V_X^c(S_{t+1}) \frac{\partial X_{t+1}}{\partial G_t} + V_B^c(S_{t+1}) \frac{\partial B_{t+1}}{\partial G_t} + V_{\Xi}^c(S_{t+1}) \frac{\partial \Xi_{t+1}}{\partial G_t} \right], \quad (\text{A.6})$$

and

$$\theta(G_t; S_t) \equiv B_t \left(- \frac{\partial s_t^g}{\partial G_t} \right). \quad (\text{A.7})$$

Assume the strengthened local second-order/regularity condition

$$H_G(G_t^*, \mu_t; S_t) < 0, \quad (\text{A.8})$$

Applying the implicit function theorem to (A.5) yields

$$\frac{\partial G_t^*}{\partial \mu_t} = - \frac{H_{\mu}(G_t^*, \mu_t; S_t)}{H_G(G_t^*, \mu_t; S_t)} = \frac{1 - \theta_t}{H_G(G_t^*, \mu_t; S_t)}, \quad (\text{A.9})$$

where $\theta_t \equiv \theta(G_t^*; S_t)$ is evaluated at the optimum. Because the denominator is negative by (A.8), the sign of $\partial G_t^* / \partial \mu_t$ is the opposite of the sign of $1 - \theta_t$.

This proves Proposition 7. □

A.2. Proof of Corollary 5: Tipping and local multiplicity

The local equilibrium choice of adaptation solves the fixed-point problem

$$G_t = \mathcal{T}(G_t; S_t) \equiv \mathcal{G}(\mathcal{M}(B_{t+1} - \bar{B}(\Xi_t)); S_t). \quad (\text{A.10})$$

Differentiating (A.10) with respect to G_t and applying the chain rule gives, at a local fixed point G_t^* ,

$$\mathcal{T}_G(G_t^*; S_t) = \frac{\partial \mathcal{G}(\mu_t; S_t)}{\partial \mu_t} \cdot \mathcal{M}'(\cdot) \cdot \frac{\partial B_{t+1}}{\partial G_t}. \quad (\text{A.11})$$

Using (30),

$$\frac{\partial B_{t+1}}{\partial G_t} = 1 + B_t \frac{\partial s_t^g}{\partial G_t}, \quad (\text{A.12})$$

so

$$|\mathcal{T}_G(G_t^*; S_t)| = |\partial \mathcal{G}(\mu_t; S_t) / \partial \mu_t| |\mathcal{M}'(\cdot)| |1 + B_t (\partial s_t^g / \partial G_t)|. \quad (\text{A.13})$$

Therefore condition (32) is exactly the statement that

$$|\mathcal{T}_G(G_t^*; S_t)| > 1. \quad (\text{A.14})$$

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This proves Corollary 5. □

B. Additional Empirical Results

This appendix reports control-set variants for the debt-change and single-crossing kink specifications. The main text treats the without-theta-controls kink model with $\mathbf{Z}_{it}$ controls as the preferred specification. The no-theta-controls kink specification displays the clearest single-crossing pattern. Specifications with direct theta controls preserve the positive low-theta effect but attenuate the high-theta debt-improving effect, suggesting that the evidence is strongest when $\widehat{\theta}_{it}^F$ is used as the sorting index rather than absorbed directly as an additional control.

Table B.1: Debt-Change Model Control-Set Variants

Panel A. Baseline debt-change model			
Dependent variable		$B_{i,t+1} - B_{it}$	
Control set	$\mathbf{Z} + B + X$	$B + X$	None
G_{it}	-0.024 (-0.569)	-0.069 (-1.513)	-0.030 (-0.577)
B_{it}	-0.093*** (-3.839)	-0.098*** (-4.607)	
X_{it}	-0.143 (-1.241)	-0.492*** (-4.306)	
Controls $\mathbf{Z}_{it}$	Yes	No	No
Observations	1,060	1,060	1,060
Adjusted R^2	0.449	0.372	0.327
Panel B. Continuous theta interaction model			
Dependent variable		$B_{i,t+1} - B_{it}$	
Control set	$\mathbf{Z} + B + X$	$B + X$	None
G_{it}	-0.016 (-0.370)	-0.056 (-1.166)	0.010 (0.182)
$G_{it} \times \hat{\theta}_{it}^F$	-0.250 (-0.576)	-0.432 (-1.009)	-1.236*** (-3.504)
B_{it}	-0.086*** (-2.675)	-0.087*** (-3.183)	
X_{it}	-0.118 (-0.974)	-0.457*** (-3.981)	
Controls $\mathbf{Z}_{it}$	Yes	No	No
Observations	1,060	1,060	1,060
Adjusted R^2	0.449	0.373	0.346
Panel C. Full-theta dynamics model			
Dependent variable		$B_{i,t+1} - B_{it}$	
Control set	$\mathbf{Z} + B + X$	$B + X$	None
G_{it}	-0.021 (-0.473)	-0.077 (-1.615)	0.038 (0.728)
$G_{it} \times \hat{\theta}_{it}^F$	-0.133 (-0.267)	0.062 (0.138)	-0.302 (-0.674)
$\hat{\theta}_{it}^F$	-0.044 (-0.584)	-0.188*** (-2.727)	-0.194*** (-3.320)
B_{it}	-0.076** (-2.369)	-0.035 (-1.130)	
X_{it}	-0.146 (-1.088)	-0.538*** (-4.311)	
Controls $\mathbf{Z}_{it}$	Yes	No	No
Observations	1,060	1,060	1,060
Adjusted R^2	0.449	0.383	0.367

Notes: Each specification includes country and year fixed effects. Robust t -statistics are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$.

For the $\mathbf{Z}_{it} + B_{it} + X_{it}$ version reported in Table B.2, the without-theta-controls kink specification is

$$\begin{aligned} \Delta B_{i,t+1} = & \alpha_i + \tau_t + aG_{it}(c - \hat{\theta}_{it}^F)_+ + bG_{it}(\hat{\theta}_{it}^F - c)_+ \\ & + \delta_B B_{it} + \delta_X X_{it} + \mathbf{\Gamma}' \mathbf{Z}_{it} + \varepsilon_{it}. \end{aligned} \quad (\text{B.1})$$

Table B.2: Single-Crossing Kink without Theta Controls: Control-Set Variants

Dependent variable Control set	$\mathbf{Z} + B + X$	$B_{i,t+1} - B_{it}$ $B + X$	None
$G_{it}(c - \hat{\theta}_{it}^F)_+$	3.402**	3.583	4.343**
$p(a > 0)$	0.010	0.052	0.005
$G_{it}(\hat{\theta}_{it}^F - c)_+$	-0.154	-0.345	-1.061***
$p(b < 0)$	0.360	0.205	0.001
RSS cutoff	0.004	-0.003	0.006
Sign-consistent	Yes	Yes	Yes
Country FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Observations	1,060	1,060	1,060
Adjusted R^2	0.452	0.375	0.351

Notes: These variants keep the main without-theta-controls kink structure but vary the additional controls. The preferred main-text cutoff remains the $\mathbf{Z}_{it}$ -controls specification. Reported p -values are one-sided tests for the sign restriction. *** $p < 0.01$, ** $p < 0.05$.

For the $\mathbf{Z}_{it} + B_{it} + X_{it}$ version corresponding to Table B.3, the kink specification with direct theta controls is

$$\begin{aligned} \Delta B_{i,t+1} = & \alpha_i + \tau_t + aG_{it}(c - \hat{\theta}_{it}^F)_+ + bG_{it}(\hat{\theta}_{it}^F - c)_+ \\ & + \rho_1 \hat{\theta}_{it}^F + \rho_2 (\hat{\theta}_{it}^F - c)_+ + \delta_B B_{it} + \delta_X X_{it} + \mathbf{\Gamma}' \mathbf{Z}_{it} + \varepsilon_{it}. \end{aligned} \quad (\text{B.2})$$

Table B.3: Single-Crossing Kink with Theta Controls

Dependent variable	$B_{i,t+1} - B_{it}$
$G_{it}(c - \hat{\theta}_{it}^F)_+$	3.560*** (3.163)
$G_{it}(\hat{\theta}_{it}^F - c)_+$	-0.113 (-0.233)
Direct theta controls	Yes
Cutoff c	0.004
RSS	1.889
Share below cutoff	0.350
Selection rule	RSS minimum; sign-consistent
Controls $\mathbf{Z}_{it}$	Yes
Country FE	Yes
Year FE	Yes
Observations	1,060
Adjusted R^2	0.452

Notes: This appendix version includes $\hat{\theta}_{it}^F$ and $(\hat{\theta}_{it}^F - c)_+$ as direct controls. The low-theta coefficient remains positive and significant, but the high-theta debt-improving coefficient is much weaker. Robust t -statistics are reported in parentheses. *** $p < 0.01$.

For the $\mathbf{Z}_{it} + B_{it} + X_{it}$ column reported in Table B.4, the estimated with-theta-controls variant uses the same control-augmented kink equation:

$$\begin{aligned}
\Delta B_{i,t+1} = & \alpha_i + \tau_t + aG_{it}(c - \hat{\theta}_{it}^F)_+ + bG_{it}(\hat{\theta}_{it}^F - c)_+ \\
& + \rho_1 \hat{\theta}_{it}^F + \rho_2 (\hat{\theta}_{it}^F - c)_+ + \delta_B B_{it} + \delta_X X_{it} + \mathbf{\Gamma}' \mathbf{Z}_{it} + \varepsilon_{it}.
\end{aligned} \tag{B.3}$$

Table B.4: Single-Crossing Kink with Theta Controls: Control-Set Variants

Dependent variable Control set	$\mathbf{Z} + B + X$	$B_{i,t+1} - B_{it}$ $B + X$	None
$G_{it}(c - \hat{\theta}_{it}^F)_+$	3.303*	2.576	5.873***
$p(a > 0)$	0.044	0.075	0.000
$G_{it}(\hat{\theta}_{it}^F - c)_+$	-0.101	0.018	-0.020
$p(b < 0)$	0.416	0.516	0.482
RSS cutoff	-0.012	-0.004	-0.004
Sign-consistent	Yes	No	Yes
Direct theta controls	Yes	Yes	Yes
Country FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Observations	1,060	1,060	1,060
Adjusted R^2	0.456	0.401	0.387

Notes: These variants keep direct theta controls and vary the additional controls across $\mathbf{Z}_{it} + B_{it} + X_{it}$, $B_{it} + X_{it}$, and no controls. Reported p -values are one-sided tests for the sign restriction. *** $p < 0.01$, * $p < 0.10$.

C. Alternative Empirical Results Using Level Readiness and Vulnerability

This appendix repeats the empirical exercise in Section 4 using level ND-GAIN readiness and vulnerability measures instead of the GDP-adjusted residual-performance measures used in the main specification. The purpose is to assess whether the reduced-form spread, debt-dynamics, and single-crossing patterns are sensitive to this alternative proxy construction. The empirical workflow, sample restrictions, fixed effects, and standard-error convention are otherwise unchanged.

C.1. Data, Proxies, and Measurement Units

The sample is the same unbalanced non-U.S. country-year panel covering 1995–2023. The dependent variable in the spread regressions is the sovereign spread, s_{it} . The adaptation-readiness proxy is now

$$G_{it} = 0.01 \times readiness100_{it}, \quad (\text{C.1})$$

and the vulnerability proxy is

$$X_{it} = 0.01 \times vulnerability100_{it}. \quad (\text{C.2})$$

Debt exposure remains debt/GDP, $B_{it} = 0.01 \times debt_gdp_{it}$, and sovereign spreads are scaled as $s_{it} = 0.01 \times bond_spreads_{it}$.

The control vector is denoted by $\mathbf{Z}_{it} = (Z_{1,it}, Z_{2,it}, \dots, Z_{8,it})'$. Its components

are real GDP, real GDP growth, CPI inflation, overall balance/GDP, international reserves, government effectiveness, regulatory quality, and terms of trade. All controls are scaled as in Section 4. Table C.1 reports the post-preprocessing descriptive statistics under the level-proxy construction.

Table C.1: Post-Preprocessing Descriptive Statistics, Level-Proxies

Block	Variable	N	Mean	SD	Median	Min.	Max.
S	Sovereign spread	1,272	0.024	0.043	0.007	-0.034	0.343
G	Readiness level	1,798	0.496	0.143	0.492	0.179	0.807
B	Debt/GDP	1,712	0.586	0.342	0.522	0.039	2.610
X	Vulnerability level	1,798	0.382	0.078	0.367	0.251	0.581
Z_1	Real GDP	1,791	0.080	0.029	0.077	0.019	0.163
Z_2	Real GDP growth	1,787	0.034	0.036	0.035	-0.145	0.246
Z_3	CPI inflation	1,791	0.057	0.108	0.032	-0.018	1.973
Z_4	Overall balance/GDP	1,697	-0.004	0.034	-0.005	-0.300	0.234
Z_5	International reserves	1,753	0.055	0.128	0.018	0.000	1.527
Z_6	Government effectiveness	1,550	0.006	0.009	0.007	-0.013	0.025
Z_7	Regulatory quality	1,550	0.006	0.008	0.007	-0.013	0.023
Z_8	Terms of trade	1,595	1.006	0.187	0.994	0.319	2.731

Notes: Statistics use the post-preprocessing non-U.S. 1995–2023 panel. Each row is computed on nonmissing observations for the corresponding scaled variable.

C.2. Adaptation and Sovereign Financing Costs

The baseline spread regression is identical to equation (33), except that G_{it} and X_{it} are the level readiness and vulnerability proxies defined in equations (C.1)–(C.2). Table C.2 shows that the readiness coefficient remains negative and statistically significant. The coefficient on public debt is positive and significant, while the level vulnerability coefficient is close to zero in the baseline fixed-effects specification.

Table C.2: Baseline Fixed-Effects Estimates, Level-Proxies

Dependent variable	10-year sovereign spread, s_{it}
G_{it}	-0.055*** (-3.209)
B_{it}	0.045*** (6.480)
X_{it}	0.005 (0.073)
Controls $\mathbf{Z}_{it}$	Yes
Country FE	Yes
Year FE	Yes
Sample period	1995–2023
Countries	60
Observations	1,120
Adjusted R^2	0.901

Notes: Robust t -statistics are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

C.3. Heterogeneous Spread Relief and the Empirical Debt-Improving Index

The heterogeneity regressions follow the same specifications as equations (34)–(35). The empirical marginal spread relief and debt-improving index remain

$$\hat{m}_{it}^{G,F} = -\frac{\partial \hat{s}_{it}}{\partial G_{it}}, \quad \hat{\theta}_{it}^F = B_{it} \hat{m}_{it}^{G,F}. \quad (\text{C.3})$$

Under the level-proxy construction, the debt interaction is negative and statistically significant. This indicates stronger spread relief from readiness at higher debt levels. The $G_{it} \times X_{it}$ interaction is small in the parsimonious full-interaction model but becomes positive and statistically significant when $G_{it} \times \mathbf{Z}_{it}$ interactions are added, so the vulnerability-level interaction should be interpreted as proxy-specific rather than as direct evidence on vulnerability-performance sorting.

Table C.3: Heterogeneous Spread Relief, Level-Proxies

Dependent variable	10-year sovereign spread, s_{it}			
G_{it}	0.069*** (3.233)	-0.062 (-0.803)	0.063 (0.815)	-0.166 (-1.498)
B_{it}	0.196*** (8.760)	0.045*** (6.469)	0.196*** (8.749)	0.186*** (8.186)
X_{it}	0.074 (1.067)	-0.002 (-0.018)	0.068 (0.560)	-0.257* (-1.936)
$G_{it} \times B_{it}$	-0.276*** (-7.734)		-0.276*** (-7.734)	-0.261*** (-7.192)
$G_{it} \times X_{it}$		0.016 (0.091)	0.014 (0.074)	0.835*** (3.911)
Controls $\mathbf{Z}_{it}$	Yes	Yes	Yes	Yes
$G_{it} \times \mathbf{Z}_{it}$ controls	No	No	No	Yes
Country FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Countries	60	60	60	60
Observations	1,120	1,120	1,120	1,120
Adjusted R^2	0.914	0.900	0.914	0.918

Notes: Columns 1–4 report debt heterogeneity, vulnerability-level heterogeneity, full interaction, and full interaction with $G_{it} \times \mathbf{Z}_{it}$, respectively. Robust t -statistics are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table C.4: Empirical Debt-Improving Index Descriptive Statistics, Level-Proxies

Variable	N	Mean	SD	Median	Min.	Max.
$\hat{\theta}_{it}^F$	1,120	0.070	0.153	0.028	-0.158	1.235
$\hat{m}_{it}^{G,F}$	1,120	0.065	0.108	0.065	-0.382	0.540
$\partial \hat{s}_{it} / \partial G_{it}$	1,120	-0.065	0.108	-0.065	-0.540	0.382

Notes: Statistics are computed on the expanded full-interaction spread-model estimation sample. The marginal relief variable is $\hat{m}_{it}^{G,F} = -\partial \hat{s}_{it} / \partial G_{it}$, and the empirical debt-improving index is $\hat{\theta}_{it}^F = B_{it} \hat{m}_{it}^{G,F}$.

C.4. Visualizing Spread Relief and the Empirical Index

Figure C.1 plots the marginal spread relief implied by the level-proxy full-interaction specification. Panel A varies debt/GDP while holding the vulnerability proxy at its median. Panel B varies the vulnerability proxy while holding debt/GDP at its

median. The confidence intervals are delta-method intervals based on the robust variance-covariance matrix of $(\hat{\beta}_G, \hat{\beta}_{GB}, \hat{\beta}_{GX})$.

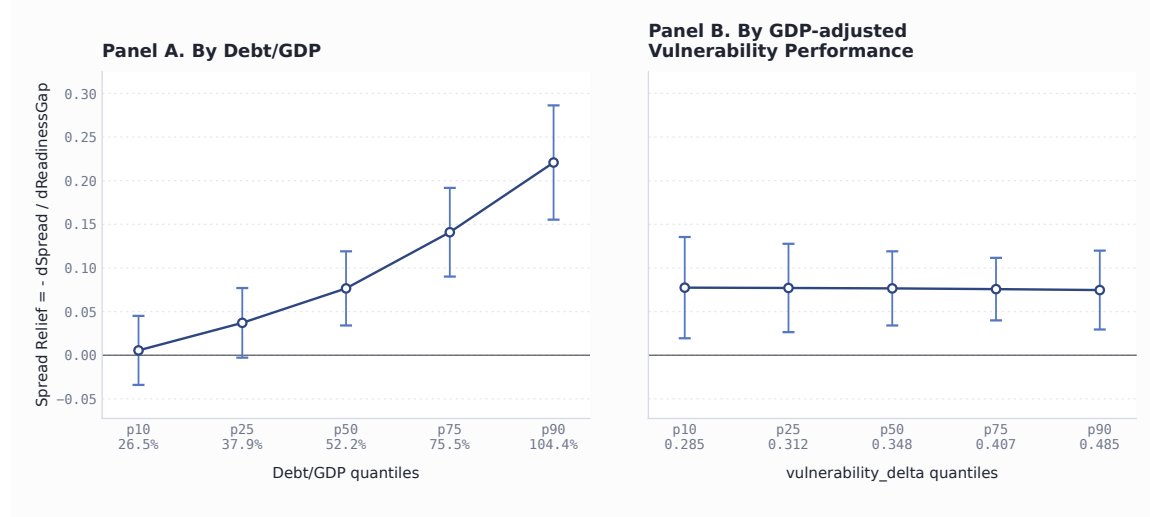


Figure C.1: Marginal Spread Relief from the Full-Interaction Spread Model, Level-Proxies
Notes: Panel A fixes X_{it} at its median. Panel B fixes B_{it} at its median. This appendix varies the level vulnerability proxy rather than the GDP-adjusted vulnerability-performance measure used in the main specification.

Figure C.2 plots the country-year distribution of $\hat{\theta}_{it}^F$ and country-level average rankings under the level-proxy construction. The dashed line marks the preferred empirical cutoff selected by the without-theta-controls kink specification.

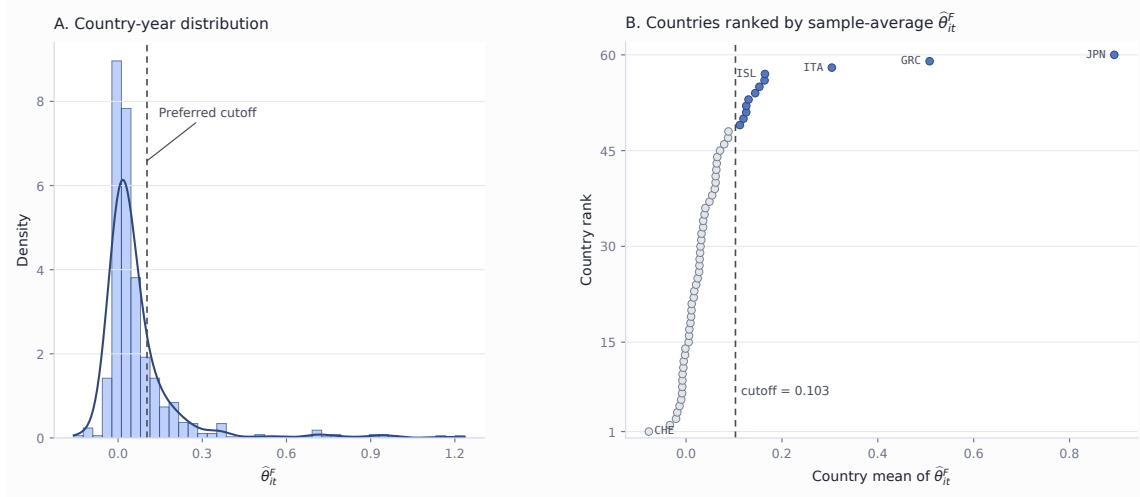


Figure C.2: Distribution of the Empirical Debt-Improving Index, Level-Proxies

Notes: The left panel plots the country-year distribution of $\hat{\theta}_{it}^F$. The right panel ranks countries by sample-average $\hat{\theta}_{it}^F$. The dashed line marks the preferred empirical cutoff from the without-theta-controls kink specification.

C.5. Debt Dynamics and the Continuous Theta Interaction

The debt-change regressions follow equations (38)–(40), using the level-proxy version of $\hat{\theta}_{it}^F$. Table C.5 shows that the continuous theta interaction remains negative and statistically significant. This implies that observations with higher empirical debt-improving capacity have a lower marginal effect of readiness on next-period debt changes.

Table C.5: Debt-Change Regressions, Level-Proxies

Dependent variable	$\Delta B_{i,t+1} = B_{i,t+1} - B_{it}$		
G_{it}	-0.103*** (-2.614)	-0.082** (-2.074)	-0.090** (-2.118)
$G_{it} \times \hat{\theta}_{it}^F$		-0.277*** (-3.619)	0.033 (0.057)
$\hat{\theta}_{it}^F$			-0.187 (-0.520)
Real GDP	8.171*** (5.412)	4.694*** (3.205)	4.523*** (2.914)
Real GDP growth	-0.326*** (-3.610)	-0.369*** (-4.180)	-0.391*** (-3.854)
CPI inflation	-0.166** (-1.976)	-0.166** (-2.027)	-0.161** (-2.095)
Overall balance/GDP	-0.468*** (-5.849)	-0.448*** (-5.572)	-0.436*** (-5.160)
International reserves	0.020 (1.465)	-0.007 (-0.475)	-0.005 (-0.350)
Government effectiveness	-1.913** (-2.020)	-1.174 (-1.222)	-1.111 (-1.134)
Regulatory quality	1.921** (2.155)	0.331 (0.358)	-0.029 (-0.025)
Terms of trade	0.044*** (2.774)	0.035** (2.355)	0.036** (2.500)
Theta main effect	No	No	Yes
Controls $\mathbf{Z}_{it}$	Yes	Yes	Yes
Country FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Observations	1,060	1,060	1,060
Countries	60	60	60
Adjusted R^2	0.425	0.443	0.444

Notes: Columns 1–3 report the baseline model, the continuous theta interaction model, and the full-theta dynamics model, respectively. The full-theta dynamics specification includes $\hat{\theta}_{it}^F$ as a direct control. Robust t -statistics are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

C.6. Grouped Debt-Change Heterogeneity

Table C.6 compares theta, debt-ratio, and marginal-relief sorts under the level-proxy construction. The theta and marginal-relief panels show more negative coefficients in higher groups, while the debt-ratio sort is less monotone. This is consistent with the interpretation that the composite index $B_{it}\hat{m}_{it}^{G,F}$ captures more than debt exposure alone.

Table C.6: Grouped Debt-Change Heterogeneity Regressions, Level-Proxies

Panel A. Full-theta groups					
Dependent variable	$B_{i,t+1} - B_{it}$				
Group	0–20%	20–40%	40–60%	60–80%	80–100%
G_{it}	0.008 (0.045)	-0.001 (-0.017)	-0.234*** (-2.803)	-0.133 (-1.139)	-0.417 (-1.622)
Controls $\mathbf{Z}_{it}$	Yes	Yes	Yes	Yes	Yes
Country FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Observations	212	212	212	212	212
Countries	23	35	41	38	24
Adjusted R^2	0.597	0.550	0.412	0.489	0.576
Panel B. Debt-ratio groups					
Dependent variable	$B_{i,t+1} - B_{it}$				
Group	0–20%	20–40%	40–60%	60–80%	80–100%
G_{it}	0.033 (0.533)	-0.157 (-1.169)	-0.164** (-2.220)	-0.167 (-1.328)	-0.604** (-2.443)
Controls $\mathbf{Z}_{it}$	Yes	Yes	Yes	Yes	Yes
Country FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Observations	212	212	212	212	212
Countries	28	35	41	32	22
Adjusted R^2	0.465	0.482	0.705	0.397	0.629
Panel C. Marginal-relief groups					
Dependent variable	$B_{i,t+1} - B_{it}$				
Group	0–20%	20–40%	40–60%	60–80%	80–100%
G_{it}	0.037 (0.223)	0.045 (0.598)	-0.161* (-1.821)	-0.316*** (-3.129)	-0.434 (-1.426)
Controls $\mathbf{Z}_{it}$	Yes	Yes	Yes	Yes	Yes
Country FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Observations	212	212	212	212	212
Countries	25	34	41	38	27
Adjusted R^2	0.576	0.610	0.413	0.697	0.588

Notes: Each column estimates the next-period change in debt/GDP on the level readiness proxy, G_{it} , within the indicated subgroup. Macro control coefficients and subgroup cutoff values are omitted for compactness. Robust t -statistics are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

C.7. Single-Crossing Kink Test

The single-crossing specification again estimates whether the marginal effect of readiness on next-period debt changes crosses zero as a function of $\widehat{\theta}_{it}^F$. For a candidate cutoff c , the estimated without-theta-controls model is

$$\Delta B_{i,t+1} = \alpha_i + \tau_t + aG_{it}(c - \widehat{\theta}_{it}^F)_+ + bG_{it}(\widehat{\theta}_{it}^F - c)_+ + \mathbf{\Gamma}'\mathbf{Z}_{it} + \varepsilon_{it}. \quad (\text{C.4})$$

The implied marginal effect is

$$m(\theta; c) = a(c - \theta)_+ + b(\theta - c)_+. \quad (\text{C.5})$$

Table C.7 selects an empirical cutoff of $c = 0.103037$ in the main without-theta-controls specification. Below the cutoff, the estimated marginal effect is positive; above the cutoff, it is negative. The sign pattern is therefore consistent with the single-crossing interpretation under the level-proxy construction.

Table C.7: Single-Crossing Kink Marginal-Effect Model, Level-Proxies

Dependent variable	$B_{i,t+1} - B_{it}$
$G_{it}(c - \hat{\theta}_{it}^F)_+$	0.726*** (3.667)
$G_{it}(\hat{\theta}_{it}^F - c)_+$	-0.225*** (-2.890)
Cutoff c	0.103
RSS	1.912
Share below cutoff	0.810
Selection rule	RSS minimum; sign-consistent
Controls $\mathbf{Z}_{it}$	Yes
Country FE	Yes
Year FE	Yes
Observations	1,060
Adjusted R^2	0.447

Notes: The exact RSS-selected cutoff is $c = 0.1030368341448663$. The model includes country and year fixed effects and $\mathbf{Z}_{it}$ controls, but does not include direct theta controls. Robust t -statistics are reported in parentheses. *** $p < 0.01$.

Figure C.3 plots the continuous marginal-effect curve implied by equation (C.5). The confidence band is computed by the delta method using the robust variance-covariance matrix of the two kink coefficients.

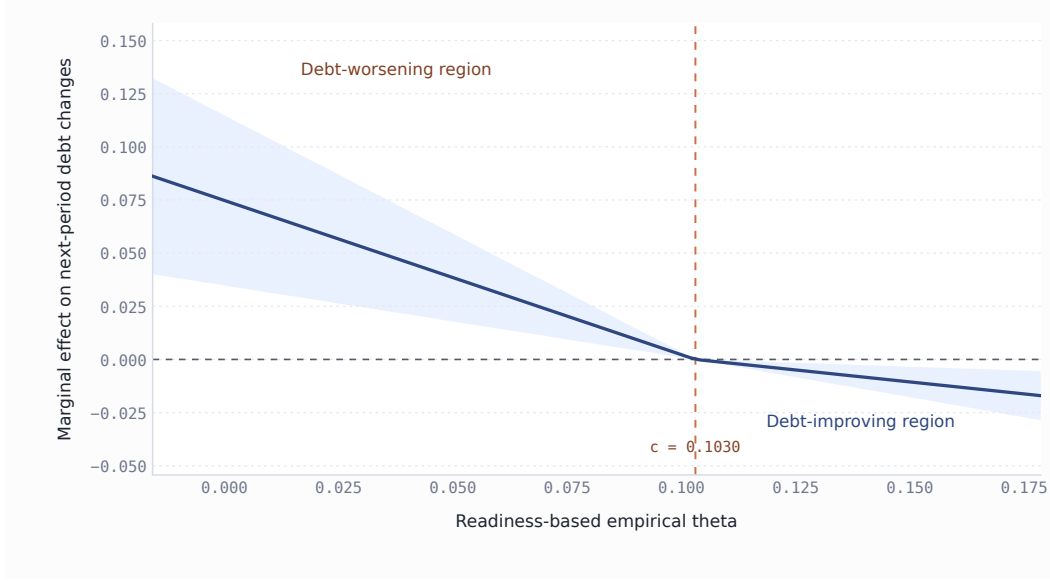


Figure C.3: Continuous Kink Marginal Effect without Theta Controls, Level-Proxies
Notes: The figure plots $m(\theta; c) = a(c - \theta)_+ + b(\theta - c)_+$ from the single-crossing kink model without theta controls. Positive values imply that readiness improvements are associated with higher next-period debt changes; negative values imply lower next-period debt changes. The vertical dashed line marks the empirical cutoff c .

C.8. Country Screens Around the Preferred Cutoff

Table C.8 applies the preferred cutoff to country-level screens. Under the level-proxy construction, no country satisfies the low-debt, always-above-cutoff, top-relief screening rule. Four countries satisfy the high-debt, always-below-cutoff, bottom-relief rule. As in the main text, the screen is descriptive and should not be interpreted as a structural country classification.

Panel A. Low debt, theta always above the cutoff, and top-quartile marginal relief

Table C.8: Country Screens Around the Preferred Cutoff, Level-Proxies

Country	ISO3	Sample period	Years	Mean debt/GDP	Mean theta	Mean $\hat{m}^{G,F}$
No countries satisfy the screening rule.						

Panel B. High debt, theta always below the cutoff, and bottom-quartile marginal relief

Country	ISO3	Sample period	Years	Mean debt/GDP	Mean theta	Mean $\hat{m}^{G,F}$
Croatia	HRV	2008–2023	16	69.66%	-0.021	-0.034
Malta	MLT	2010–2023	14	53.91%	-0.014	-0.028
Namibia	NAM	2014–2023	10	53.74%	-0.008	-0.026
Netherlands	NLD	2000–2023	23	54.21%	-0.007	-0.016

Notes: The preferred cutoff is $c = 0.103037$. Screening thresholds use country-level means: the mean-debt median is 53.06 percent, the bottom quartile of mean $\hat{m}_{it}^{G,F}$ is 0.009, and the top quartile is 0.102. The always-above and always-below conditions are evaluated year by year.

C.9. Additional Control-Set Variants

This subsection reports the appendix control-set variants generated by the same workflow. The models parallel Appendix B, but use the level readiness and vulnerability proxies throughout.

Table C.9: Debt-Change Model Control-Set Variants, Level-Proxies

Panel A. Baseline debt-change model			
Dependent variable		$B_{i,t+1} - B_{it}$	
Control set	$\mathbf{Z} + B + X$	$B + X$	None
G_{it}	-0.066 (-1.612)	-0.125*** (-2.898)	-0.137*** (-3.145)
B_{it}	-0.085*** (-3.514)	-0.084*** (-3.951)	
X_{it}	0.323 (1.315)	-0.278 (-1.276)	
Controls $\mathbf{Z}_{it}$	Yes	No	No
Observations	1,060	1,060	1,060
Adjusted R^2	0.450	0.362	0.332
Panel B. Continuous theta interaction model			
Dependent variable		$B_{i,t+1} - B_{it}$	
Control set	$\mathbf{Z} + B + X$	$B + X$	None
G_{it}	-0.068* (-1.664)	-0.133*** (-3.132)	-0.139*** (-3.289)
$G_{it} \times \hat{\theta}_{it}^F$	-0.100 (-0.905)	-0.232** (-2.015)	-0.332*** (-4.858)
B_{it}	-0.065* (-1.808)	-0.034 (-0.949)	
X_{it}	0.343 (1.399)	-0.142 (-0.666)	
Controls $\mathbf{Z}_{it}$	Yes	No	No
Observations	1,060	1,060	1,060
Adjusted R^2	0.450	0.369	0.368
Panel C. Full-theta dynamics model			
Dependent variable		$B_{i,t+1} - B_{it}$	
Control set	$\mathbf{Z} + B + X$	$B + X$	None
G_{it}	-0.069 (-1.646)	-0.141*** (-2.979)	-0.150*** (-3.116)
$G_{it} \times \hat{\theta}_{it}^F$	-0.075 (-0.131)	-0.061 (-0.113)	-0.052 (-0.095)
$\hat{\theta}_{it}^F$	-0.016 (-0.047)	-0.111 (-0.348)	-0.165 (-0.507)
B_{it}	-0.064** (-2.199)	-0.028 (-0.909)	
X_{it}	0.339 (1.260)	-0.160 (-0.721)	
Controls $\mathbf{Z}_{it}$	Yes	No	No
Observations	1,060	1,060	1,060
Adjusted R^2	0.450	0.368	0.368

Notes: Each specification includes country and year fixed effects. Robust t -statistics are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

For the $\mathbf{Z}_{it} + B_{it} + X_{it}$ version reported in Table C.10, the without-theta-controls kink specification is

$$\begin{aligned} \Delta B_{i,t+1} = & \alpha_i + \tau_t + aG_{it}(c - \hat{\theta}_{it}^F)_+ + bG_{it}(\hat{\theta}_{it}^F - c)_+ \\ & + \delta_B B_{it} + \delta_X X_{it} + \mathbf{\Gamma}' \mathbf{Z}_{it} + \varepsilon_{it}. \end{aligned} \quad (\text{C.6})$$

Table C.10: Single-Crossing Kink without Theta Controls: Control-Set Variants, Level-Proxies

Dependent variable Control set	$\mathbf{Z} + B + X$	$B_{i,t+1} - B_{it}$ $B + X$	None
$G_{it}(c - \hat{\theta}_{it}^F)_+$	1.078*	-0.202	0.141
$p(a > 0)$	0.050	0.783	0.240
$G_{it}(\hat{\theta}_{it}^F - c)_+$	-0.118	-0.188	-0.339***
$p(b < 0)$	0.147	0.050	0.000
RSS cutoff	-0.002	0.042	-0.016
Sign-consistent	Yes	No	Yes
Country FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Observations	1,060	1,060	1,060
Adjusted R^2	0.451	0.366	0.364

Notes: These variants keep the main without-theta-controls kink structure but vary the additional controls. Reported p -values are one-sided tests for the sign restriction. *** $p < 0.01$, * $p < 0.10$.

For the $\mathbf{Z}_{it} + B_{it} + X_{it}$ version corresponding to Table C.11, the kink specification with direct theta controls is

$$\begin{aligned} \Delta B_{i,t+1} = & \alpha_i + \tau_t + aG_{it}(c - \hat{\theta}_{it}^F)_+ + bG_{it}(\hat{\theta}_{it}^F - c)_+ \\ & + \rho_1 \hat{\theta}_{it}^F + \rho_2 (\hat{\theta}_{it}^F - c)_+ + \delta_B B_{it} + \delta_X X_{it} + \mathbf{\Gamma}' \mathbf{Z}_{it} + \varepsilon_{it}. \end{aligned} \quad (\text{C.7})$$

Table C.11: Single-Crossing Kink with Theta Controls, Level-Proxies

Dependent variable	$B_{i,t+1} - B_{it}$
$G_{it}(c - \hat{\theta}_{it}^F)_+$	0.093 (0.251)
$G_{it}(\hat{\theta}_{it}^F - c)_+$	0.470 (0.745)
Direct theta controls	Yes
Cutoff c	0.179
RSS	1.896
Share below cutoff	0.900
Selection rule	RSS minimum
Controls $\mathbf{Z}_{it}$	Yes
Country FE	Yes
Year FE	Yes
Observations	1,060
Adjusted R^2	0.450

Notes: This appendix version includes $\hat{\theta}_{it}^F$ and $(\hat{\theta}_{it}^F - c)_+$ as direct controls. The with-theta-controls version does not satisfy the single-crossing sign pattern at the RSS-minimizing cutoff. Robust t -statistics are reported in parentheses.

For the $\mathbf{Z}_{it} + B_{it} + X_{it}$ column reported in Table C.12, the estimated with-theta-controls variant uses the same control-augmented kink equation in (C.7).

Table C.12: Single-Crossing Kink with Theta Controls: Control-Set Variants, Level-Proxies

Dependent variable	$B_{i,t+1} - B_{it}$		
Control set	$\mathbf{Z} + B + X$	$B + X$	None
$G_{it}(c - \hat{\theta}_{it}^F)_+$	1.614**	2.007***	1.495**
$p(a > 0)$	0.020	0.004	0.018
$G_{it}(\hat{\theta}_{it}^F - c)_+$	0.141	0.234	0.232
$p(b < 0)$	0.593	0.660	0.656
RSS cutoff	0.044	0.044	0.046
Sign-consistent	No	No	No
Direct theta controls	Yes	Yes	Yes
Country FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Observations	1,060	1,060	1,060
Adjusted R^2	0.454	0.373	0.367

Notes: These variants keep direct theta controls and vary the additional controls across $\mathbf{Z}_{it} + B_{it} + X_{it}$, $B_{it} + X_{it}$, and no controls. Reported p -values are one-sided tests for the sign restriction. *** $p < 0.01$, ** $p < 0.05$.